

Entity Identification Report

Project Title

Online Food Delivery Management System

Objective

The objective of this project is to identify the entities, attributes, primary keys, and foreign keys required for designing the database of an Online Food Delivery Management System.

Business Scenario

An online food delivery company requires a centralized database to manage customers, restaurants, food items, orders, payments, deliveries, and customer reviews. The system should reduce data redundancy, improve order processing, and provide accurate information for business operations.

Entities and Attributes

Customer

- CustomerID (Primary Key)
- Name
- Email
- MobileNumber
- Address
- RegistrationDate

Restaurant

- RestaurantID (Primary Key)
- RestaurantName
- Address
- ContactNumber

FoodItem

- FoodItemID (Primary Key)
- FoodName
- Price
- AvailableQuantity
- RestaurantID (Foreign Key)

Order

- OrderID (Primary Key)
- CustomerID (Foreign Key)
- OrderDate
- TotalAmount
- OrderStatus

OrderDetails

- OrderDetailID (Primary Key)
- OrderID (Foreign Key)
- FoodItemID (Foreign Key)
- Quantity
- UnitPrice

Payment

- PaymentID (Primary Key)
- OrderID (Foreign Key)
- PaymentMethod
- PaymentDate
- PaymentStatus

DeliveryPartner

- DeliveryPartnerID (Primary Key)
- Name
- MobileNumber
- VehicleNumber

Delivery

- DeliveryID (Primary Key)
- OrderID (Foreign Key)
- DeliveryPartnerID (Foreign Key)
- DeliveryAddress
- DeliveryStatus

Review

- ReviewID (Primary Key)
- CustomerID (Foreign Key)
- FoodItemID (Foreign Key)
- Rating
- Comments

Relationship Types

One-to-One (1:1)

- Order → Payment
- Order → Delivery

One-to-Many (1:M)

- Customer → Order
- Restaurant → FoodItem
- Order → OrderDetails
- DeliveryPartner → Delivery
- Customer → Review
- FoodItem → Review

Many-to-One (M:1)

- Order → Customer
- FoodItem → Restaurant
- OrderDetails → Order
- OrderDetails → FoodItem
- Delivery → DeliveryPartner

Many-to-Many (M:N)

Order ↔ FoodItem

This relationship is implemented using the OrderDetails table.

Self-Reference

Not applicable for this project because none of the entities have a recursive relationship.

Cardinality

- Customer (1) → Order (M)
- Restaurant (1) → FoodItem (M)
- Order (1) → OrderDetails (M)
- FoodItem (1) → OrderDetails (M)
- Order (1) → Payment (1)
- Order (1) → Delivery (1)
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Conclusion

The identified entities and attributes provide the foundation for designing the database schema of the Online Food Delivery Management System.